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Separation of telechelic oligomers according to architecture by liquid chromatography

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ABSTRACT

Telechelic oligomers based on diethylene glycol or lower polyethylene glycols and caprolactone or butene oxide can be separated according to their architecture by means of liquid chromatography under adsorption conditions for the hydrophobic block and critical or exclusion conditions for PEG. This behavior, which is predicted by the theory, can be verified by the experiment for block copolymers of ethylene oxide with ε-caprolactone and butene oxide. The individual peaks were identified by matrix-assisted time-of-flight mass spectrometry (MALDI-TOF-MS) of fractions obtained by semipreparative liquid chromatography at critical conditions of center block (PEG) and adsorption conditions for outer blocks (PCL and PBO).

1. Introduction

Telechelic polymers and oligomers with terminal hydroxy or amino groups can serve as prepolymers for the production of polyurethanes, as is the case with Jeffamines (EO–PO copolymers with terminal amino groups) and caprolactone-diols.

Such polymers are typically produced by (anionic) ring opening polymerization of epoxides or lactones with a difunctional starting material, which may be uniform or polydisperse (often a monomeric diol or oligomers or polymers of ethylene oxide or propylene oxide).

If the polymerization proceeds in both directions, the resulting polymer has a triblock structure. Such samples contain, however, always some amounts of the homopolymer (due to chain transfer reactions) and the diblock, which is formed, if the polymerization proceeds in just one direction.

Obviously, di- and triblocks may show quite different physical properties, such as phase behavior, melting point etc. It has been shown, that the physical properties of EO-PO diblocks are quite different from that of EO-PO-EO or PO-EO-PO triblocks. The micellization behavior [1–6] and the influence on protein adsorption [7] of above mentioned di- and triblock copolymers are quite different from each other. Other amphiphilic block copolymers also show similar behavior, as has been shown experimentally [8,9] and by Monte Carlo simulations [10].

Of course the triblocks may also differ in symmetry: the arms of a triblock may have equal or different length: the highest degree of asymmetry corresponds to the diblock, and there may be symmetric and asymmetric triblocks with different degree of asymmetry (Table 1).

The monomers of outer block (if more than one on one side) can arrange themselves in either head to head or head to tail positions. In the case of racemic epoxides there may also be diastereomeric sequences.

The characterization of such block copolymers is an extremely difficult task. The techniques, which may be utilized, are liquid chromatography using different separation mechanisms and mass spectrometry (mainly MALDI-TOF-MS). Obviously, MALDI-TOF-MS provides information on molar mass distribution (MMD) and functionality, but the spectra of highly polydisperse samples are very complicated and difficult to interpret. Moreover, MALDI-TOF-MS is not capable of discriminating molecules with the same molar mass, but different architecture.

In principle, a combination of a chromatographic separation with spectroscopy (MS or NMR) would be the most promising approach.

Some years ago, Chang and coworkers reported a separation of di- and triblock copolymers consisting of polyethylene oxide (PEO) and poly L-lactide (PLLA) at critical conditions of PEO and

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Table 1
Possible structures in A_nBA_m block copolymers, and monoisotopic masses of corresponding diols derived from ε-caprolactone or butane oxide (A) and diethylene glycol (B).

n + m	N^a	Diblock	Triblock					Monoisotopic mass/Da (CL-DEG-CL)	Monoisotopic mass/Da (BO-DEG-BO)
		Asymmetric	Symmetric	Asymmetric					
1	1	AB	–	–	–	–	–	220.13	178.24
2	2	A_2B	ABA	–	–	–	–	334.20	250.36
3	2	A_3B	–	ABA_2	–	–	–	448.27	322.48
4	3	A_4B	A_2BA_2	ABA_3	–	–	–	562.34	394.60
5	3	A_5B	–	ABA_4	A_2BA_3	–	–	676.40	466.72
6	4	A_6B	A_3BA_3	ABA_5	A_2BA_4	–	–	790.47	538.84
7	4	A_7B	–	ABA_6	A_2BA_5	A_3BA_4	–	904.54	610.96
8	5	A_8B	A_4BA_4	ABA_7	A_2BA_6	A_3BA_5	–	1018.61	683.08
9	5	A_9B	–	ABA_8	A_2BA_7	A_3BA_6	A_4BA_5	1132.68	755.20
10	6	$A_{10}B$	A_5BA_5	ABA_9	A_2BA_8	A_3BA_7	A_4BA_6	1246.74	827.32

N^a: Number of possible structures.

adsorption conditions of PLLA. They were able to achieve separation of block copolymers according to symmetry [11,12]. The same group demonstrated the effect of block architectures on elution behavior in liquid chromatography at critical conditions when non-critical block is excluded from the pores of the stationary phase [13].

A similar approach also appeared to be suitable for the purpose of this study. There is, however, a serious problem, if the center block is rather short, as will be explained later on. Hence we have tried to find chromatographic conditions, which allow a discrimination of telechelic oligomers of this type according to symmetry.

For this purpose, we have chosen samples with oligomers of ethylene oxide as center block (mainly diethylene glycol, DEG) and caprolactone or butene oxide as monomeric unit. These products could have the structures and molar masses given in Table 1.

2. Separation mechanisms in liquid chromatography of polymers

The elution of a molecule in liquid chromatography is commonly described by the following relation:

$$V_e = V_i + K \cdot V_p \tag{1}$$

wherein V_e is the elution volume, V_i is the interstitial volume, V_p is the pore volume, and K is the distribution coefficient – a parameter which characterizes the partitioning of the molecule between the flowing and the stagnant part of the liquid phase.

The theory of chromatography of polymers is well developed [14–16]. The different regimes can be characterized by the so-called interaction parameter c, which describes the interaction of a structural unit with the stationary phase [17].

2.1. Homopolymers

The most common technique in liquid chromatography of polymers is size exclusion chromatography (SEC), which separates according to molecular dimensions.

In SEC, c is strongly negative, and K can assume values between 0 and 1. A polymer elutes in between V_i and $V_i + V_p$; elution volumes in SEC decrease with increasing molar mass M [14].

SEC may be used to determine the overall MMD, but it does not yield any information on architecture.

Liquid adsorption chromatography (LAC) separates according to the number of structural units, which may interact with the surface of the stationary phase, and the strength of this interaction.

In LAC, c is positive, $K > 1$ and increases with molar mass M. Consequently, the elution volumes are typically much larger than the void volume $V_0 = V_i + V_p$.

In a special mobile phase composition (at the critical point of adsorption) the distribution coefficient K will be unity, and a polymer will elute at the void volume $V_0 = V_i + V_p$ [15,16].

2.2. Diblocks and monofunctional polymers

If a molecule contains different structural units (repeat units or end groups), a quite complicated peak pattern may be obtained.

In block structures containing two different repeat units (of A and B), two interaction parameters (c_A and c_B) must be taken into account. Obviously, there may be the following situations:

1 c_A and c_B are negative: the block copolymer elutes in SEC mode.
2 c_A and c_B are positive: the polymer elutes in LAC mode.
3 $c_A = 0$; $c_B < 0$: A is invisible, separation according to B in SEC mode.
4 $c_A = 0$; $c_B > 0$: A is invisible, separation according to B in LAC mode.
5 $c_A > 0$; $c_B < 0$. The copolymers AB are strongly retained, like in the LAC mode, and the individual homologues of B elute in SEC order.

In the analysis of AB diblocks, a highly useful technique is liquid chromatography under critical conditions (LCCC), which is the limiting case between size exclusion and adsorption: at the critical adsorption point (CAP) polymer homologues of different molar mass elute at the same position. At the CAP for A (situations 3 and 4) the corresponding block becomes chromatographically invisible, and a diblock copolymer AB can be separated according to the block B.

The behavior of a diblock in LCCC is very simple [18,19]. At the CAP for B, a diblock copolymer AB will elute in a narrow peak with the same elution volume as the homopolymer A: $V_{AB} = V_A = V_0 + V_pK_A$.

For monofunctional molecules (with one end group a), the distribution coefficient $K^{(a)}$ at the CAP for B is given by

$$K^{(a)} \approx 1 + q_a \tag{2}$$

wherein q_a is the effective interaction parameter for the end-group a [20,21].

In AB diblocks a similar relation holds for each A oligomer. Evidently, the exact equation for AB diblocks is equivalent to the equation for a monofunctional polymer. If the interaction condition for the block A is of a strong attractive type ($c_A > 1/R_A$), the effective q_a parameter for a copolymer can be expressed as a function of both interactive and molecular parameters of the block A:

$$q_a \approx \frac{2}{c_A d}\left[\exp({c_A}^2{R_A}^2) - \frac{c_A R_A}{\sqrt{\pi}}\right] \approx \frac{2\exp({c_A}^2{R_A}^2)}{c_A d} \tag{3}$$

In situation 4, retention increases with the number of A units like in LAC. All chains with a given n umber of A units elute in a narrow peak regardless of their number of B units.

Situation 5 is utilized in liquid exclusion adsorption chromatography (LEAC) [22], which is especially useful in the analysis of functional oligomers, where B is uniform. Under such conditions, the individual homologues of *B* elute in SEC order, but far behind the void volume. The behavior of diblocks in LEAC has been theoretically described in Refs. [22,23], and was studied experimentally by Trathnigg and Gorbunov [22]. According to the papers, the characteristic feature of this mode is that the retention order by the length of the block *B* is similar to that in SEC, but diblocks are strongly retained, like in LAC.

2.3. Triblocks or difunctional oligomers

For diblocks, situations 4 and 5 yield rather simple peak patterns, the elution behavior of triblocks is, however, much more complicated. While ABA triblocks at the CAP for A behave more or less like AB diblocks. At the CAP for *B*, the B block in an *ABA* triblock does not become invisible (as it would do in an *AB* diblock) [21]. A similar behavior can be observed in the case of difunctional chains with adsorbing end groups A: this can even be utilized in the analysis of difunctional polymers: under such conditions a separation according to the length of the block *B* can be achieved (quite similar to LEAC) [20,24]. According to the theory the distribution coefficient $K^{(a,a)}$ for symmetrical difunctionals is equal to

$$K^{(a,a)} \approx 1 + 2q_a + q_a^2 \cdot \frac{d}{\sqrt{\pi} \cdot R} \tag{4}$$

wherein *d* is the pore diameter of the stationary phase and *R* the radius of gyration of the critical block. Evidently, symmetrical difunctionals are equivalent to symmetric triblocks with uniform A blocks.

Obviously, this effect is very pronounced for chains with a short center block, and it will be negligible for chains with a long critical block (large *R*) in narrow pores (small *d*) and rather weak interaction of the end group: in this case $q^2 \cdot \frac{d}{\sqrt{\pi} \cdot R} \to 0$.

For asymmetrical difunctionals – with different end groups – the distribution coefficient is given by

$$K^{(a1,a2)} \approx 1 + q_{a1} + q_{a2} + q_{a1} \cdot q_{a2} \cdot \frac{d}{\sqrt{\pi} \cdot R} \tag{5}$$

Obviously, difunctionals with different end groups correspond to asymmetric triblocks with uniform A blocks and different number of repeat units.

This offers a chance to discriminate di- and triblocks with the same number of repeat units in the *A* blocks: at the CAP for *B* and adsorption conditions for *A*, the triblock should elute earlier than the diblock.

A serious problem may arise, if the center block is rather short and polydisperse. In this case, it will not be chromatographically invisible even under critical conditions. This may result in overlapping series of peaks, which cannot be resolved.

This problem does not arise for samples with a uniform center block. In this case there is no need to keep strictly critical conditions. In mobile phases corresponding to LEAC conditions a considerably better separation according to symmetry can be expected.

As follows from the theory [23], the behavior of triblocks in LEAC is qualitatively similar to that of diblocks, but the retention turns dependent of the architecture of block copolymers. In particular, theory has predicted different retention in LEAC of diblocks, symmetric and asymmetric triblocks which are similar in all aspects apart from the architecture. Accordingly, dependences of the retention on the length of the block *B* under LEAC conditions are more pronounced than at the CAP. Therefore one may expect that the separation of block copolymers by the architecture should be possible in LEAC mode, provided that the blocks are monodisperse or possess narrow molar mass distributions. In LEAC, separation by architecture increases with the increase of adsorption of blocks *A*; better separations can be expected in wide pores, at $0.1 < R_B/d < 0.5$.

3. Experimental

3.1. Materials

ε-Caprolactone diols, polyethylene glycols, butene oxide and ε-Caprolactone were purchased from Fluka (Buchs, Switzerland). The triblocks were synthesized by microwave-assisted polymerization of ε-Caprolactone using tin (II) octoate catalyst [25] and anionic ring opening polymerization of butene oxide [26]. Oligoethylene glycols (uniform such as diethylene glycol, triethylene glycol, tetraethylene glycol or polydisperse oligoethylene glycols such as PEG 200 and PEG 300) were used as initiator for both types of polymerizations. Special focus were products formed using DEG as initiator. The HPLC grade solvents (water, acetone, acetonitrile, and methanol) for chromatography were purchased from Carl Roth Gmbh, Karlsruhe, Germany.

3.2. Chromatography

Three different chromatographic systems were used for this study.

Preparative chromatograph consists of JASCO 880 PU pump (Japan Spectrosopic Company, Tokyo, Japan), Rheodyne 7125 injection valve (Rheodyne, Cotati, CA, USA) equipped with a 500 μL loop and a semi-preparative silica based reversed phase octadecyl column: Spherisorb ODS2 (25 mm × 300 mm, 5 μm, 80 Å) from PhaseSep (Deeside, UK). The detection system for preparative chromatography consists of density detection system DDS70 (CHROMTECH, Graz, Austria) combined with a SICON LCD 201 RI detector. An Advantec SF 2120 fraction collector (Advantec, Dublin, CA, USA) was used for fraction collection. A thermostat Lauda RM6 (Lauda-Königshofen, Germany) was used to maintain the temperature of the thermostated box at 25.0 °C, in which columns and density cell were placed. All preparative separations were performed at flow rate of 2.0 mL/min. The mobile phase used for caprolactone polymers was acetonitrile–water whereas methanol–water for butene oxide polymers (compositions are given in figure captions). CHROMA software package was used for data acquisition and processing.

Mobile phases for preparative chromatography were mixed by mass and vacuum degassed. A DMA 60 density meter equipped with a measuring cell DMA 602M (A. Paar, Graz, Austria) was used to control the composition of mobile phase by density measurement.

Some analytical investigations were performed on a chromatographic system consists of a JASCO 880 PU pump (from Japan Spectrosopic Company, Tokyo, Japan), Rheodyne 7125 injection valve (from Rheodyne, Cotati, CA, USA) equipped with an 50 μL loop and a dual detection system. The dual detection system in this case consists of density detector DDS70 (CHROMTECH, Graz, Austria) connected to a Waters 2414 refractive index detector (Waters, Milford, MA, USA). CHROMA software package (which has been developed for DDS70) was used for data acquisition and processing.

The gradient separations in the second dimensions were performed on a gradient system Ultimate 3000 (Dionex, Germerink, Germany). This system consists of a pump DGP-3600A, solvent degasser SRD-3600, column thermostat TCC-3000, and an autosampler WPS-3000SL. The detection in gradient HPLC was conducted by an evaporative light scattering detector PL-ELS 2100 (Polymer Laboratories, Church Stretton, Shropshire, UK). Chromeleon (Dionex, Germerink, Germany) is the software package used for data acquisition and processing.

The following columns were used for analytical chromatography (specifications given by the producer):

Synergi Fusion RP (silica-based dodecyl phase; 250 mm × 4.6 mm; particle diameter = 4 μm; nominal pore size = 80 Å, from Phenomenex, Torrance, CA, USA).

Prodigy ODS3 (Phenomenex, Torrance, CA, USA): silica-based octadecyl phase; 250 mm × 4.6 mm; particle diameter = 5 μm; nominal pore size = 100 Å.

Mobile phase compositions used on prodigy column were from 40% to 60% acetonitrile–water and on synergi column from 70% to 85% methanol–water in isocratic mode while following gradients were used on synergi column to get separation of reasonable number of oligomers according to symmetry.

Gradient 1: starts at 70% methanol–water, in 30 min 80% methanol–water, in 35 min 85% methanol–water, 10 min constant at 85% methanol–water and then back to 70% methanol–water in one minute and then constant for 9 min.

Gradient 2: starts at 75% methanol–water, in 30 min 85% methanol–water, in 35 min 90% methanol–water, 10 min constant at 90% methanol–water and then back to 75% methanol–water in one minute and then constant for 9 min.

All analytical separations were performed at 25.0 °C with a flow rate of 0.5 mL/min. The mobile phases in the second dimension were mixed by the gradient pump and their compositions are given in volume per cent. The mobile phase composition for isocratic separations were mass per cent. The mobile phases used for this study are good solvents for both hydrophilic and hydrophobic segments of the products under study.

3.3. Mass spectrometry

Micromass TofSpec 2E Time-of-Flight Mass Spectrometer was used for MALDI-TOF mass spectrometry. The instrument is equipped with a nitrogen laser (337 nm wavelength, operated at a frequency of 5 Hz), and a time lag focusing unit. The irradiations slightly above the threshold laser power were used to generate ions. Accelerating voltage of 20 kV was applied to record positive ion spectra in reflectron mode. The system was externally calibrated with a suitable mixture of poly(ethylene glycol)s (PEG). To improve the signal-to-noise ratio, the spectra of 100–150 shots were averaged. Data analysis was performed with MassLynx-Software V3.5 (Micromass/Waters, Manchester, UK).

A solution of dithranol or DHB (c = 10 mg/mL in THF), a solution of the analyte (c = 2 mg/mL in THF), and a solution of CF_3COONa (NaTFA; c = 0.1 mg/mL in THF), in a ratio of 7:1:1 (v/v/v) were mixed to prepare sample solutions. 0.5 μL of the resulting mixture were deposited on the sample plate (stainless steel) and allowed to dry under air.

4. Results and discussion

As is shown in another paper, two series of peaks are observed in PCL diols on reversed phase columns under LCCC or LEAC conditions [25]. Such samples are synthesized by polymerization of ε-caprolactone with difunctional initiators (typically oligoethylene glycols or other diols), from which the chains should grow in both directions to yield polymers with a triblock structure HO-CL_m-X-CL_n-OH. It is, however, also possible, that CL is attached only to one end of the chain, thus yielding products with a diblock structure HO-X-CL_{n+m}-OH. These molecules have the same functionality (two hydroxy end groups) and the same mass, hence they cannot be discriminated by standard MALDI-TOF-MS.

But discrimination (and a separation) is possible by liquid chromatography at critical conditions (LCCC), as has been shown previously [20,21,24]. At the CAP for the block A and LAC conditions for B di- and triblocks (A–B and A–B–A) elute at the same volume, while B–A–B triblocks elute earlier, and their elution volumes triblocks will be the smaller, the larger the block A is.

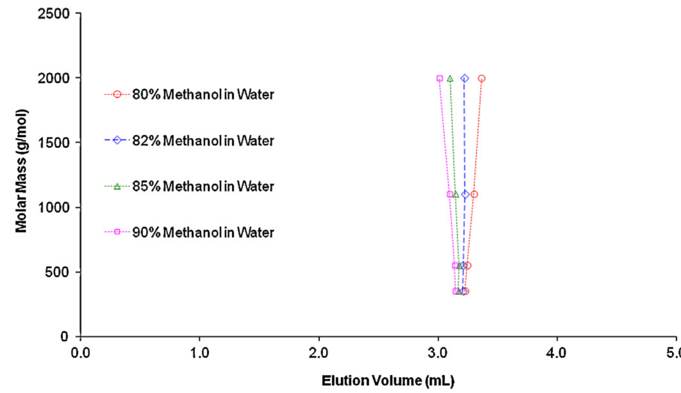

Fig. 1. Critical diagram of PEGMMEs on Synergi Fusion RP at 25 °C in methanol–water mobile phases of different composition (v/v).

While for samples with a polydisperse block A, a complex peak pattern must be expected, however, two series of peaks will be observed if A is uniform. This is indeed the case with the PCL diols based on uniform central block such as diethylene glycol (DEG), trierthylene glycol (TEG) etc. Similarly, for triblock copolymers of polybutene oxide with uniform middle block (diethylene glycol or triethylene glycol), several peaks of a single oligomer can be observed depending upon block structure (see Table 1).

In this study, triblock copolymers of ε-caprolactone and butene oxide with DEG as central block are used as model compounds to demonstrate the discrimination of block structure. The separations for both types of copolymers will be discussed separately.

4.1. Separations for CL-DEG-CL triblock copolymers

Fig. 1 shows, that on a Synergi Fusion RP column a CAP for EO exists at 25 °C in a methanol–water mobile phase containing 82 vol.% methanol. Under such conditions, two series of peaks are observed for the different PCL diols (Fig. 2): the elution volume of the individual peaks shows only a weak dependence on the length of the EO block (which is, of course, very short). Obviously, even at the CAP the peaks of the triblocks are somewhat broader in the sample based on PEG 200 (even though this is a starting material with a quite narrow MMD).

On ODS column, PEGMMEs (polyethylene glycol monomethyl ethers) show no molar mass dependence and elute at more or less the same elution volume irrespective of their molar mass in

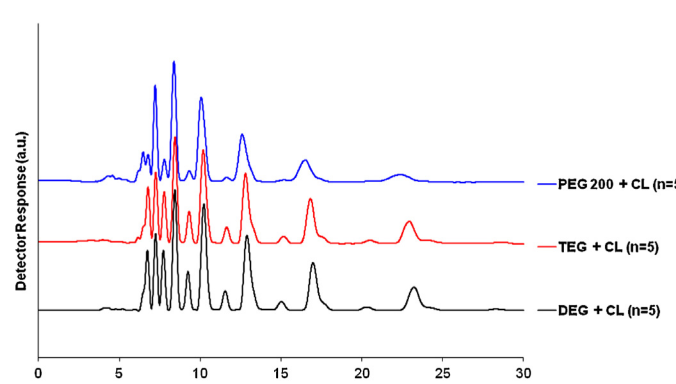

Fig. 2. LCCC of PCL-diols with a different middle block on Synergi Fusion RP in 82% methanol–water (CAP for EO).

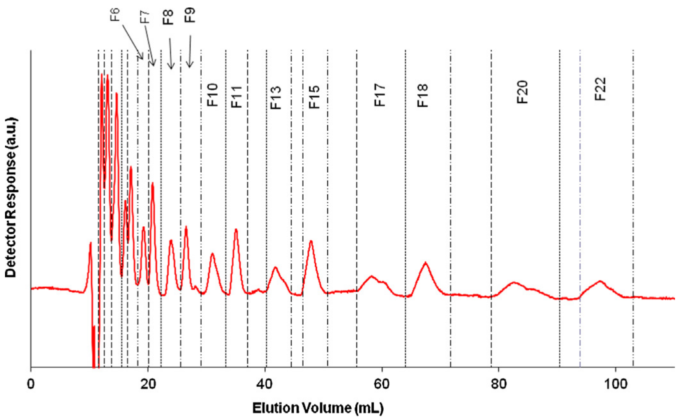

Fig. 3. Semi-preparative LCCC of PCL-diol 830 on the Spherisorb ODS2 column in 69.1% acetonitrile. Fraction limits are indicated by dashed lines.

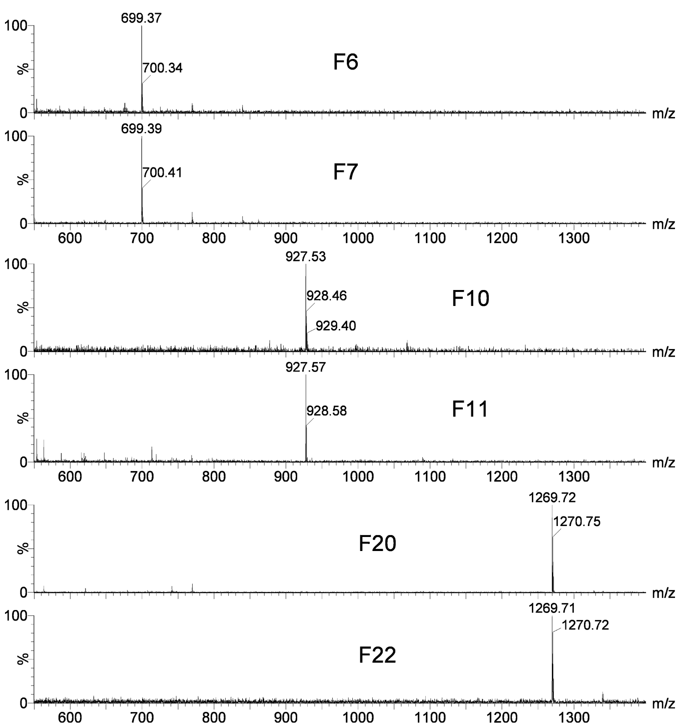

Fig. 4. MALDI-TOF mass spectra of fractions from the separation in Fig. 3 showing *pseudo*-molecular ions [M+Na]$^+$.

69.1 wt.% acetonitrile in water. On a semipreparative ODS column, we have fractionated PCL diol 830 and observed two series of peaks, one of which is sharp, while the other one is rather broad (Fig. 3).

The fractions were analyzed by MALDI-TOF-MS (Fig. 4). The fractions 6 and 7 showed only one important signal at $m/z = 699.37$ and 699.39 Da, respectively, what corresponds well to the calculated monoisotopic mass of block polymer ions [M+Na]$^+$ containing 5 CL units ($C_{34}H_{60}O_{13}Na^+$, $m/z_{calc.} = 699.39$ Da; compare also Table 1). Fractions 10 and 11 contained products with 7 CL units ($C_{46}H_{80}O_{17}Na^+$, $m/z_{calc} = 927.53$ Da), and fractions 20 and 22 species with 10 CL units ($C_{64}H_{110}O_{23}Na^+$, $m/z_{calc} = 1269.73$ Da). The spectra of the other fractions (not shown here) also corresponded to the expected mass.

These fractions were now compared to the raw samples in order to identify the peaks of the individual oligomers in LCCC (Fig. 5). It can be seen that fraction 10 and 11 (from Fig. 3) that are oligomer number 7 (see MALDI spectra, Fig. 4) are well separated from each other because of different block structures.

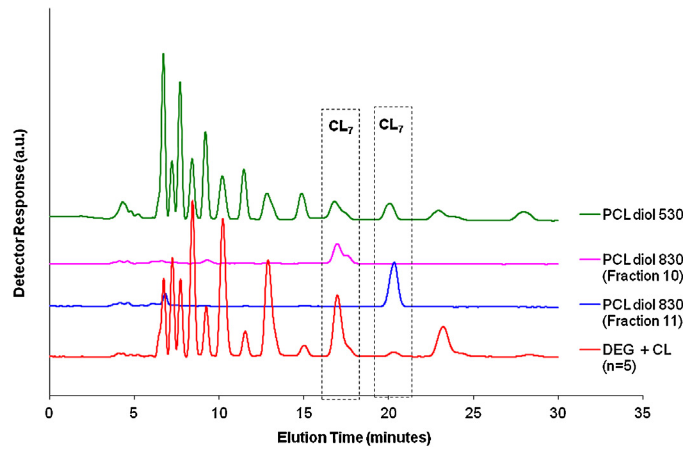

Fig. 5. LCCC of different PCL diols and two fractions from PCL diol 830 (from Fig. 3), as obtained on Synergi Fusion RP in 82% methanol–water (v/v).

In the previous paper, we have used LEAC conditions (with PCL elution in LAC mode and PEG in SEC mode) [25]. Under such conditions, a similar peak pattern is observed for the PCL diols with a uniform EO block. Two series of peak are observed, which can be identified by comparison with the fractions from LCCC. Fig. 6 depicts LEAC separation of raw CL-DEG-CL and PCL diol 530. Both samples show double peaks for all oligomers according to block structure. The oligomer numbers are assigned by MALDI-TOF-MS analysis of semipreperative LC fractions.

The resolution of peaks increases dramatically, when the mobile phase composition is changed in the direction of stronger interaction for CL, but still exclusion for EO (Fig. 7a).

In a mobile phase with 55 vol.% acetone a peak pattern is observed, which perfectly corresponds to the predicted one: in Fig. 7b, the individual peaks are labeled with their composition (wherein A denotes CL and B denotes DEG and subscript is number of CLs attached on either side of DEG).

4.2. Separations for BO-DEG-BO triblock copolymers

In principle, with monodisperse central block or a short central block with low polydispersity, the separation according to BO is possible around critical adsorption point of PEG. With uniform PEG block, there is more freedom in selection of mobile phase composition (no real need of CAP of PEG) in the analysis of BO blocks. The

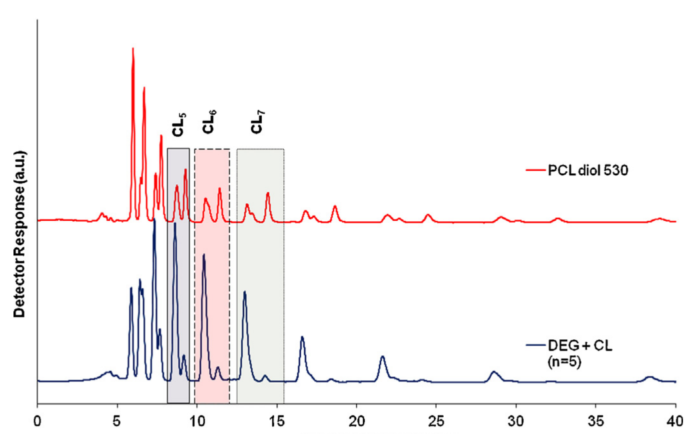

Fig. 6. LEAC of different PCL diols with assignment of peaks using the fractions of PCL diol 830 as standards (not shown), as obtained on the Synergi Fusion RP column in 70 vol.% acetone–water.

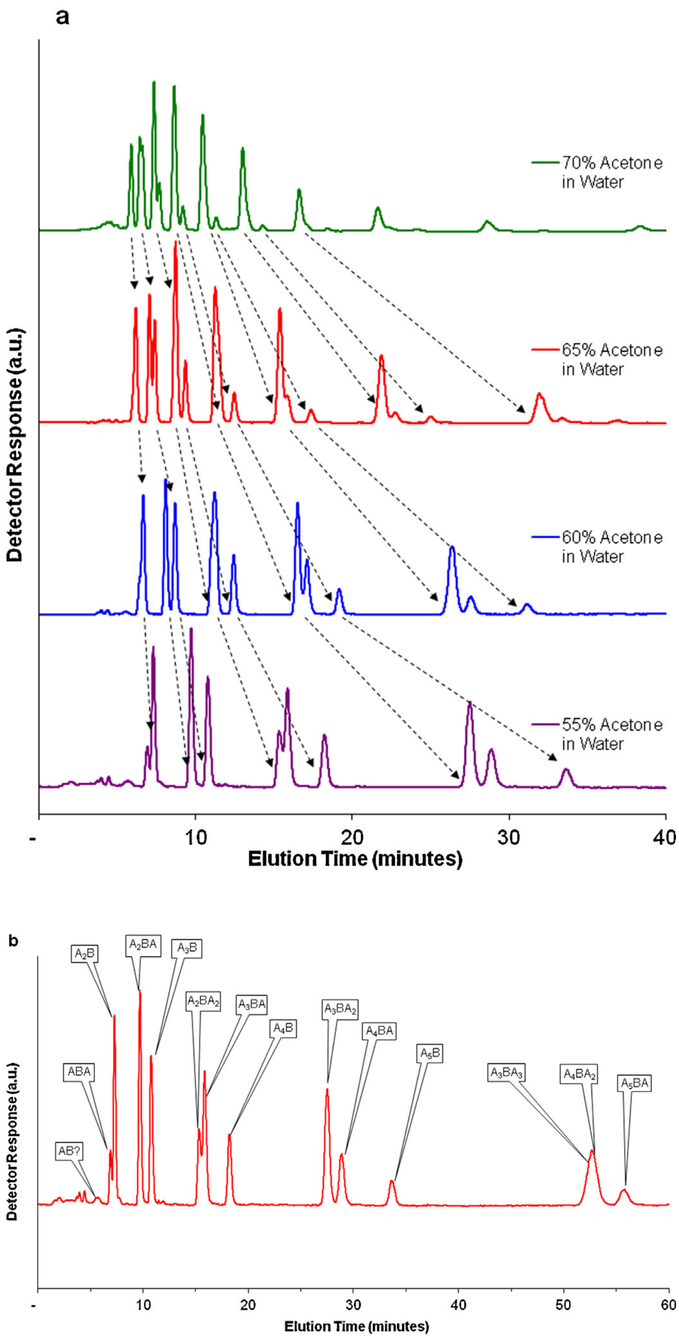

Fig. 7. (a) LEAC of the PCL-diol based on DEG with average degree of polymerization $n = 5$, as obtained on the Synergi Fusion column in acetone–water mobile phases of different composition, (b) LEAC of the PCL-diol based on DEG with average degree of polymerization $n = 5$, as obtained on the Synergi Fusion column in acetone water (55% acetone) with labeled individual peaks (B is DEG while A is CL, subscript is the number of CLs attached on either side of DEG).

oligomers of BO-DEG-BO are separated at weak interaction of BO with the stationary phase on semi-preparative scale. These fractions were collected and analyzed in second dimension at stronger interaction of BO with the stationary phase. It must be noted that in order to get good separation of all oligomers, we have to run three different products with different average number of repeat units (3, 5, and 8) in different mobile phase compositions (see Fig. 8). The fraction limits are shown by solid lines and oligomer numbers are assigned as obtained by MALDI-TOF-MS analyses of fractions (Fig. 9). The notations a,b,c and d are for further referencing. It can be seen that all the fractions are pure and uniform. The fraction a

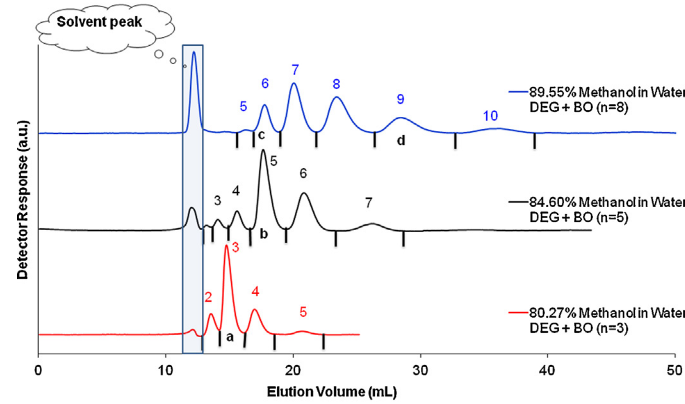

Fig. 8. Semi-preparative scale LAC separation of BO-DEG-BO oligomers according to number of BO (Oligomer number according to BO are given above each peak in different mobile phase compositions, as obtained by MALDI-TOF-MS) on Spherisorb ODS2 column (250 mm × 10 mm, 5 μm, 80 Å) from PhaseSep (Deeside, UK). Fraction limits are shown by solid lines below each peak. Notations a,b,c,d are for further fractions referencing.

corresponds to oligomer number 3, b to oligomer number 5, c to oligomer number 6 and d to oligomer number 9 with respect to BO with DEG as central segment (compare with Table 1). The spectra of the other fractions (not shown here) also corresponded to the expected mass.

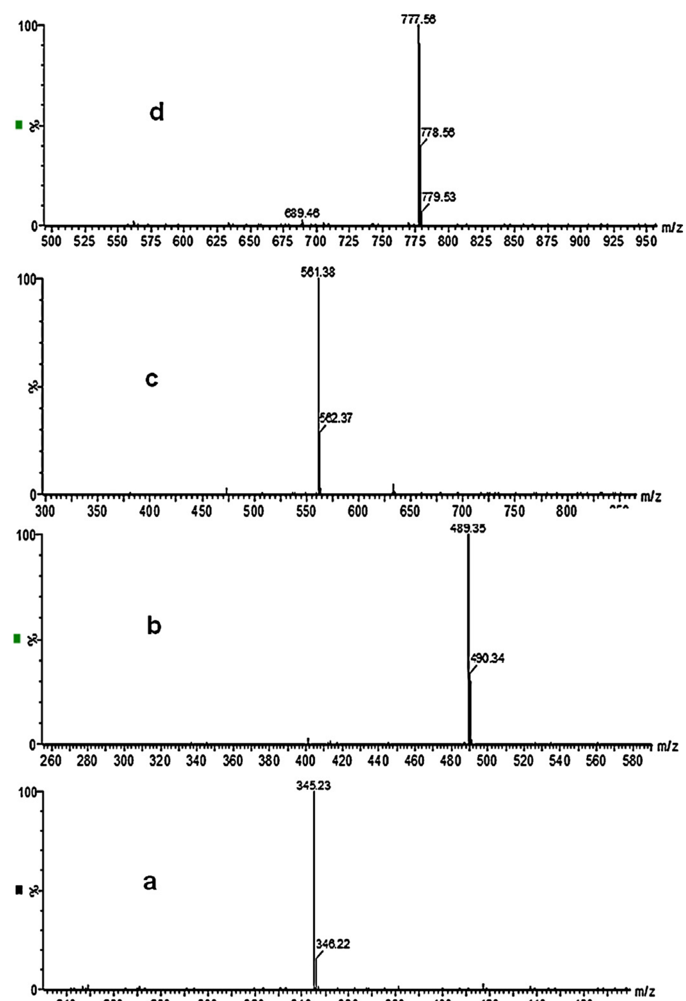

Fig. 9. MALDI-TOF-MS spectra of fractions of BO-DEG-BO showing *pseudo*-molecular ions $[M+Na]^+$, Fractions are indicated by notations a,b,c,d.

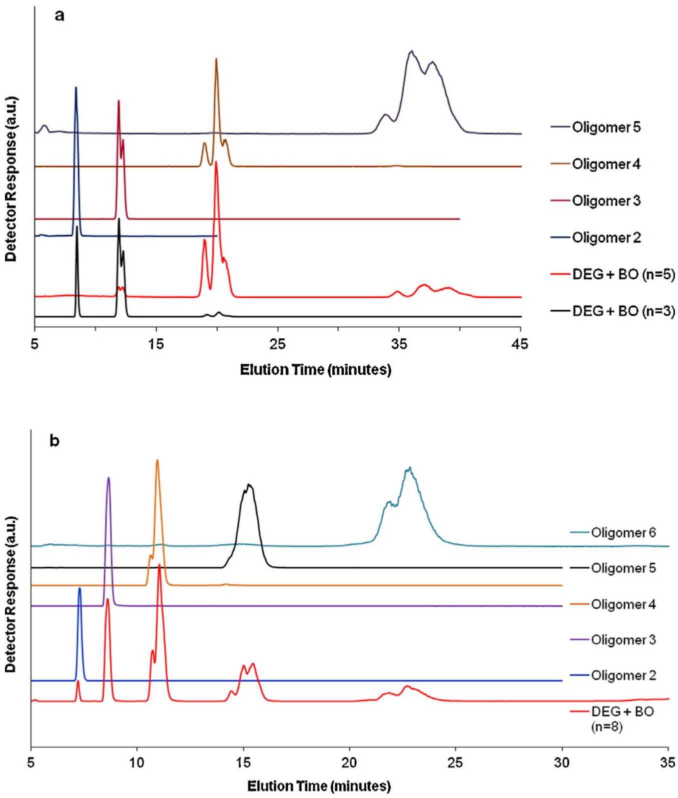

Fig. 10. Elution behavior of different BO-DEG-BO with varying degree of polymerization of BO ($n = 3, 5, 8$) and the fractions collected by semi-preparative separations (Fig. 9) on Synergi Fusion RP column in different mobile phase composition of methanol in water (a) 70%, (b) 80%.

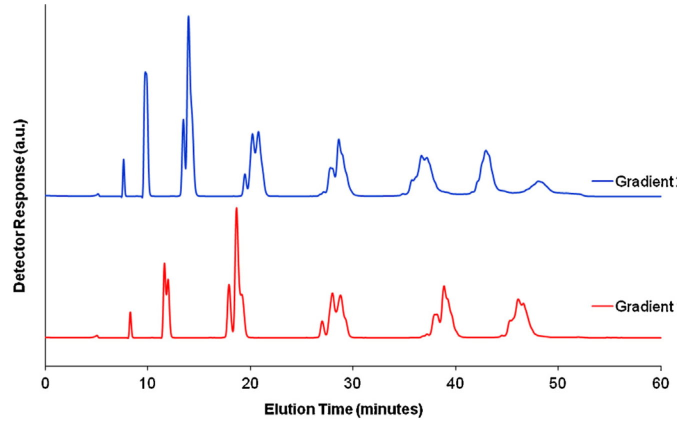

Fig. 11. Separation of BO-DEG-BO according to block structure with gradient elution on Synergi Fusion RP column in methanol–water mobile phase (details of gradients in text).

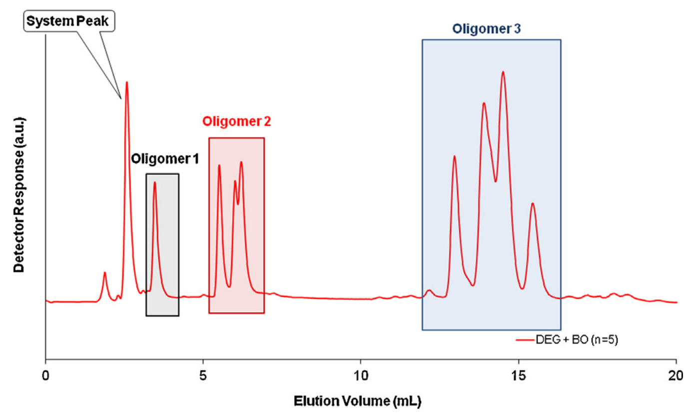

Fig. 12. Separation of lower oligomers of BO-DEG-BO on Prodigy ODS3 100 Å column in acetonitrile–water (35–65 wt.%) mobile phase.

The BO-DEG-BO triblocks can be separated according to symmetry on synergy fusion RP column in methanol–water mobile phase system. Fig. 10 shows overlays of chromatograms of raw product and collected fractions with their assigned oligomer number by MALDI-TOF-MS (see Figs. 8 and 9). It can be seen that with lower methanol content (70%), lower oligomers can be separated according to symmetry (Fig. 10a). In 80% methanol, the separation of lower oligomer according to symmetry is lost but it allows separation of oligomer number 5 and 6 with respect to block structure (Fig. 10b). With a uniform central block, the condition of being strictly on critical adsorption point is no more necessary. This allows us to run gradient elution chromatography of BO-DEG-BO polymers to achieve separation according to block structure. Gradient profile 1 starts at 70% methanol–water (see Section 2 for detailed profile), this polar mobile phase (weaker mobile phase composition for RP chromatography) leads to good separation of first six oligomers while gradient profile 2 which starts at 75% methanol–water (less polar comparatively and stronger mobile phase composition for RP chromatography compared to gradient 1) leads to separation of more oligomers (eight) but resolution of low oligomers is somewhat compromised if separation according to block structure is targeted (Fig. 11).

On reversed phase octadecyl column in acetonitrile–water mobile phase system, the chromatographic behavior is different from previous results. This system allows separation into more than expected species. Additional peaks may be explained by head to head and head to tail arrangements of BO monomers. The other possibility is diastereomers of BO block. The oligomer number 2 is resolved into three distinct peaks while oligomer number 3 is resolved into five distinct peaks (Fig. 12). The first peak of oligomer number 2 might be a symmetric product (triblock). The second and third peaks might correspond to asymmetric products (diblocks) with head to head/head to tail arrangement or their diastereomers and same is true for oligomer number 3.

5. Conclusions

A separation of di- and triblock copolymers is achieved by liquid chromatography, if the outer blocks elute in LAC mode, and the central block is short and uniform and elutes at the critical point of adsorption or in the SEC regime. Under these conditions, even a separation according to symmetry is possible. This study demonstrated the separation of diblocks from triblocks. The symmetric and asymmetric triblocks (with different degree of asymmetry) were also separated quite well from each other. The additional peaks in acetonitrile–water mobile phase on ocatdecyl RP columns might be separations according to head to head/head to tail arrangements or different tacticity of BO block. This behavior, which is predicted by the theory, is verified by the experiment.

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